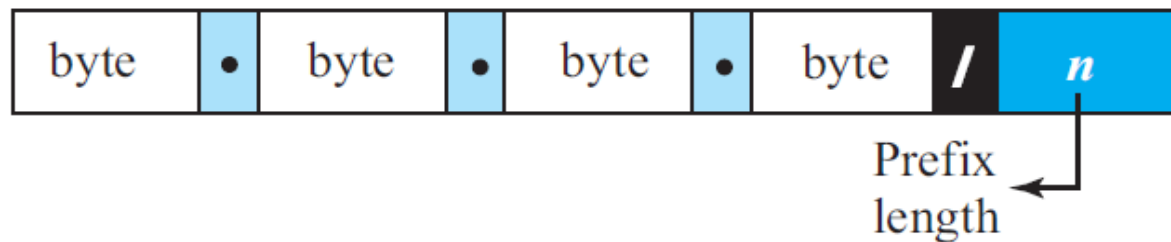


NETWORK LAYER

Module 3

Classless Addressing:

- In classless addressing, the **network** can be given a **block of addresses** of **different sizes**, depending on **how many addresses** are **actually needed**.
- Every IP addresses in classless addressing is represented as **x.y.w.z/n**, this representation is called as **CIDR** or **slash notation**. CIDR stands for **classless interdomain routing**. In this representation, **n** denotes the **prefix length**.



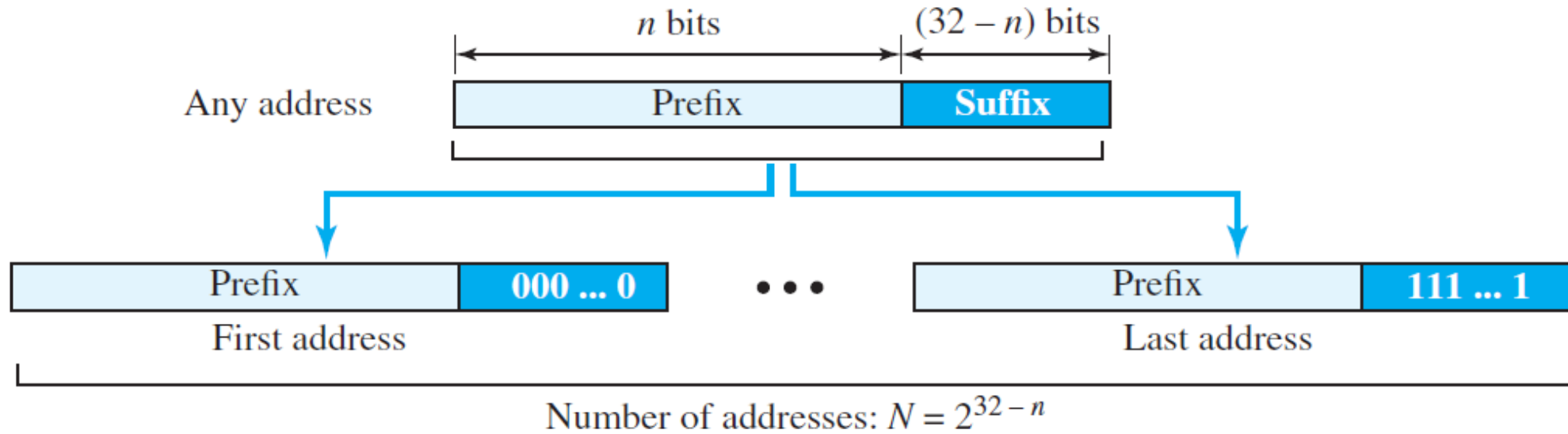
Examples:

12.24.76.8/**8**
23.14.67.92/**12**
220.8.24.255/**25**

CIDR/Slash Notation

Extracting Information from an Address:

- Given any address $x.y.w.z/n$ in the block, we can know three pieces of information about the block to which the address belongs: the number of addresses, the first address, and the last address in the block.
- Since the value of prefix length, n , is given, we can easily find these three pieces of information.
 - The number of addresses in the block is found as $N = 2^{32-n}$.
 - To find the first address, we keep the n leftmost bits and set the $(32 - n)$ rightmost bits all to 0s.
 - To find the last address, we keep the n leftmost bits and set the $(32 - n)$ rightmost bits all to 1s.



Information extraction in classless addressing

- The **first address** in the block is the **network address** and the **last address** in the block is the **broadcast address**.

Problem: A classless address is given as 167.199.170.82/27. Find the number of addresses, the first address, the last address, network address, and broadcast address in the block.

Solution: Do it in class.

Problem: A classless address is given as 167.199.170.82/27. Find the number of addresses, the first address, the last address, network address, and broadcast address in the block.

Answer: Number of Addresses = $2^{32-27} = 2^5 = 32$ addresses, First Address = 167.199.170.64/27, Last Address = 167.199.170.95/27, Network Address = First Address = 167.199.170.64/27, Broadcast Address = Last Address = 167.199.170.95/27



Problem: A classless address is given as 192.168.25.77/28. Find the number of addresses, the first address, the last address, network address, and broadcast address in the block.

Solution: Do it in class.

Problem: A classless address is given as 192.168.25.77/28. Find the number of addresses, the first address, the last address, network address, and broadcast address in the block.

Answer: Number of addresses =16, First address: 192.168.25.64/28, Last address: 192.168.25.79/28, Network address: 192.168.25.64/28, Broadcast address: 192.168.25.79/28

Address Mask:

- Another way to find the first and last addresses in the block is to use the address mask. The address mask is a 32-bit number in which the n leftmost bits are set to 1s and the rest of the bits $(32 - n)$ are set to 0s.
- We can easily find these three pieces of information:
 - The number of addresses in the block $N = \text{NOT}(\text{mask}) + 1$.
 - The first address in the block = (Any address in the block) AND (mask).
 - The last address in the block = (Any address in the block) OR [(NOT (mask))].

Problem: A classless address is given as 167.199.170.82/27. Find the mask, the number of addresses, the first address, the last address, network address, and broadcast address in the block.

Solution: Do it in class.



Problem: A classless address is given as 167.199.170.82/27. Find the number of addresses, the first address, the last address, network address, and broadcast address in the block. **Ans:** Mask = 255.255.255.224, Number of Addresses = 32 addresses, First Address = 167.199.170.64/27, Last Address = 167.199.170.95/27, Network Address = First Address = 167.199.170.64/27, Broadcast Address = Last Address = 167.199.170.95/27

References:

Data Communications and Networking by Behrouz A. Forouzan.



THANK YOU